

[Foundational Clinic 2] Grade 4-5 Powers, Exponents & Multiplication Clinic

Student Name: _____

Target:

⚡ Repeated Multiplication vs Product / Index Laws / Prime Factors of 16

Time: 35

Total:

Score: _____

Mins

100 Pts

💡 Key Conceptual Anchors:

① **Powers vs. Multiplication:** a^b means multiplying a by itself b times! $2^3 = 2 \times 2 \times 2 = 8$. It is NEVER $2 \times 3 = 6$!

② **16 has Four 2s Multiplied, NOT 8 Twos:** $16 = 2 \times 2 \times 2 \times 2 = 2^4$. $2 + 2 + \dots + 2 = 16$ is addition, not factorization!

③ **Multiplying Powers:** $a^m \times a^n = a^{m+n}$. When multiplying powers with the same base, you ADD exponents: $10^4 \times 10^5 = 10^9$ (NEVER 10^{20}).

④ **Power of Zero:** Any non-zero number to power 0 equals 1: $10^0 = 1$ (NOT 10).

Section 1: Write it Out! Expanded Form vs Simple Multiplication (3 pts each, Score: _____)

Write out the expanded multiplication form and calculate the result:

Expression	Expanded Form (e.g. $a \times a \times \dots$)	True Value	Common Slip (DO NOT MAKE)
2^3	$2 \times 2 \times 2$	8	$2 \times 3 = 6$ ❌
2^4			$2 \times 4 = 8$ ❌
3^3			$3 \times 3 = 9$ ❌
3^4			$3 \times 4 = 12$ ❌
4^2			$4 \times 2 = 8$ ❌
5^3			$5 \times 3 = 15$ ❌
10^3			$10 \times 3 = 30$ ❌

Section 2: Addition of Exponents vs Multiplication of Exponents (4 pts each, Score: _____)

1. Simplify each expression into a single power (State why exponents are added):

(a) $10^4 \times 10^5 = 10^{\text{---} + \text{---}} = \text{---}$

(b) $2^3 \times 2^4 = 2^{\text{---} + \text{---}} = \text{---}$

(c) $3^2 \times 3^5 = 3^{\text{---} + \text{---}} = \text{---}$

(d) $10^0 = \text{---}$

(e) $5^1 \times 5^2 \times 5^3 = 5^{\text{---}} = \text{---}$

(f) $10^{-3} = \frac{1}{10^{\text{---}}} = \text{---}$

Section 3: Factors vs Multiples & Square Numbers Concept (4 pts each, 24 pts) **Score:** _____

2. [Factors of 16] A student claims: "16 can be written as eight 2s."

- (a) Is the student thinking about addition or multiplication?
(b) What is the true prime factor decomposition of 16 in index notation?

- (a) The student is confusing addition with multiplication: $8 \times 2 = 16$ is $2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$.
(b) Prime factorization: $16 =$ _____ (How many 2s multiplied?)

3. [Square Numbers Verification] Which of the following numbers are square numbers? Circle them and write their twin factors:

6: Square? Yes / No

9: Square? Yes / No

18: Square? Yes / No

25: Square? Yes / No

36: Square? Yes / No

72: Square? Yes / No

Explain why 6 is NOT a square number:

Section 4: Distributive Property vs Pure Multiplication (7 pts each, 28 pts) Score: _____

4. [Distributive Law Diagnosis] In which of the following can you use the distributive property? State YES or NO and explain:

(a) $4 \times (2.5 + 7.5)$

Distributive Property applicable? _____

(b) $4 \times (2.5 \times 7.5)$

Distributive Property applicable? _____

Rule: The distributive property $a \times (b + c) = a \times b + a \times c$ applies ONLY when there is addition/subtraction inside parentheses. Pure multiplications like $4 \times (2.5 \times 7.5)$ use associative property!

5. [Two-Step Word Problem] A warehouse stores boxes in stacks. Each stack has 3^2 layers, and each layer contains 2^3 boxes.

(a) Calculate how many boxes are in one stack.

(b) If there are 5 identical stacks, find the total number of boxes.

(a) $3^2 =$ _____, $2^3 =$ _____.

Boxes in one stack = $3^2 \times 2^3 =$ _____.

(b) Total boxes in 5 stacks = _____.



[Foundational Clinic 2] Answer Key & Scoring Guide

Section 1: Expanded Form Table (24 pts)

- $2^4 = 2 \times 2 \times 2 \times 2 = 16$
- $3^3 = 3 \times 3 \times 3 = 27$
- $3^4 = 3 \times 3 \times 3 \times 3 = 81$ (NOT 12)
- $4^2 = 4 \times 4 = 16$
- $5^3 = 5 \times 5 \times 5 = 125$
- $10^3 = 10 \times 10 \times 10 = 1000$

Section 2: Exponents Product Rule (24 pts)

1. (a) $10^{4+5} = 10^9$ (NOT 10^{20}); (b) $2^{3+4} = 2^7 = 128$; (c) $3^{2+5} = 3^7$; (d) $10^0 = 1$; (e) $5^{1+2+3} = 5^6$; (f) $\frac{1}{10^3} = 0.001$.

Section 3: Factors, Multiples & Squares (24 pts)

2. (a) Addition; (b) $16 = 2 \times 2 \times 2 \times 2 = 2^4$ (four 2s multiplied).
3. Squares: 9 (3×3), 25 (5×5), 36 (6×6). Not squares: 6, 18, 72. 6 is not a square because its prime factors 2×3 are different.

Section 4: Distributive Property & Applications (28 pts)

4. (a) YES: $4 \times 2.5 + 4 \times 7.5 = 10 + 30 = 40$. (b) NO: $4 \times 2.5 \times 7.5 = 10 \times 7.5 = 75$.
5. (a) $3^2 = 9$, $2^3 = 8$. One stack = $9 \times 8 = 72$ boxes. (b) 5 stacks = $72 \times 5 = 360$ boxes.